Production systems & planning

National 5 · Materials & manufacture · Worksheet

D&M WITH MR KELLY

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ compare one-off and mass production
- ☐ explain why standard components and knock-down fittings are used
- ☐ produce a working drawing, cutting list and sequence of operations

I DO — Worked example

Read how a full-mark answer is built

1 Outline three benefits to a manufacturer of using standard components. (3 marks)

Full-mark answer each ✓ is one mark

They are cheaper to buy in bulk than making the parts in-house ✓. They are made to a consistent quality, so the manufacturer knows they will work ✓. They are readily available, so fewer production stages are needed and production is faster ✓.

Why marks are not awarded

"Easy to replace if they break."

0 marks: this is a benefit to the consumer, not the manufacturer.

WE DO — Guided practice

Use the starters and word bank to help you

A handmade oak dining table made to order.

2 Outline two reasons why one-off production is used for the table. (2 marks)

Sentence starters

One … is …

Another … is …

Word bank

methods · traditional · materials · quality · measure · craftspeople · personalised · skilled

Tip: 2 marks = 2 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) Which document lists each part with its material and sizes?
☐ Working drawing ☐ Cutting list ☐ Specification ☐ Sequence of operations

b) Which is a disadvantage of standard components for a manufacturer?
☐ Quality can't be guaranteed at all ☐ They are difficult and slow to source ☐ Dependence on an outside supplier ☐ Parts must always be made in-house

c) Which system has the highest set-up cost but the lowest cost per item?
☐ One-off ☐ Mass ☐ Prototype ☐ Hand-made

4 Key terms: write the letter of the matching description next to each term.

1. **Mass production**	____	**A** A single unique product made by highly skilled workers.
2. **One-off production**	____	**B** Large volumes of identical products; automated; low skill; low unit cost.
3. **Standard component**	____	**C** A part made in large numbers to an agreed standard (e.g. bolts, hinges).

5 Explain what is meant by ‘Cutting list’. (2 marks)

Check and reflect
Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?